REACTIVEDYNAMICS.JL — THE AGENTIC LAYER

ReactiveDynamics.jl Modeling Copilot

From a business decision that depends on how an operation behaves to a checked, reproducible model and a number you can defend, in a conversation, without anyone writing code.

THE PROBLEM

Decisions about how an operation behaves need a model, and a model needs a modeler

- A launch reservation, a bid under a staffing ceiling, a pipeline whose funding follows its results, a backlog that feeds back into demand: the answer depends on **how the operation behaves over time**.
- A spreadsheet cannot represent that. Contention, delay and feedback are interactions between rows over time, and a spreadsheet's rows are independent by construction.
- So these questions go to a specialist, take weeks, and come back as a workbook the decision-maker has to take on faith.
- Most never get a model at all.

THE VALUE

More decisions get a real model, the owner stays in the loop, the answer travels

- **A model for the cost of a conversation.** The threshold for a real model drops from "worth a specialist's quarter" to "worth an hour".
- **The owner stays in the loop.** The interview runs in business language with the person who owns the decision. Every assumption lands in a brief they signed, including which inputs were their own estimates.
- **The answer travels.** A study ends in a generated report the modeler can forward: the answer, the model, the evidence and the provenance, laid out to one template.
- **It compounds.** The next question about the same operation starts from the same model.

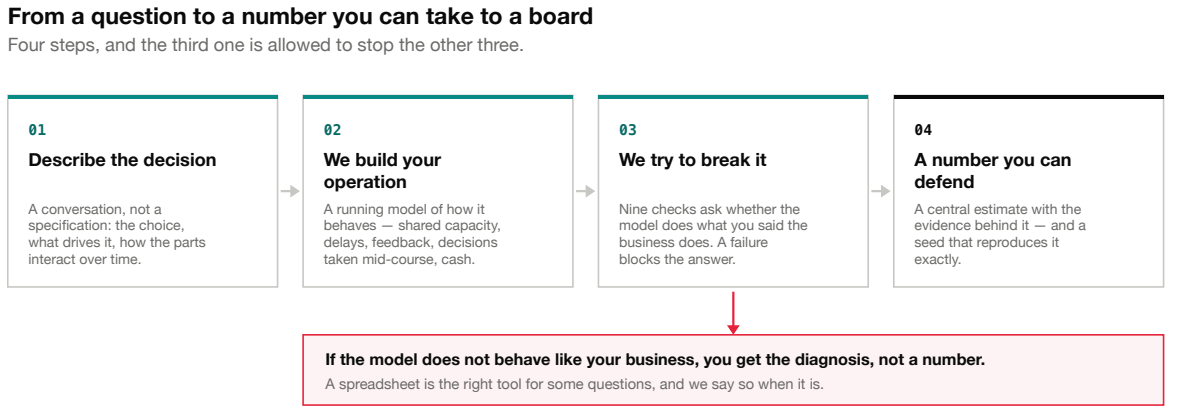
What it can model — and what the copilot checks today

Pattern	Example	Copilot today
Contention	two launches on one fill line	authored and checked end to end
Lead times, clocks	tech transfer, a build, a review	authored and checked
Feedback	backlog throttles intake; share lost after a stock-out	authored and run (12 state-dependent rates in the case); loop checks next (W25)
Mid-course decisions	review a contract at a readout	authored and run; the case turned on one
Lifecycles	projects moving through phases	authored and run
Coupled models	supply feeding commercial; several sites	in the engine (AlgebraicAgents); not yet through the copilot (W25)

HOW IT WORKS

Four steps, and the third may stop the rest

Step	You get	Gate
Frame	a signed brief	the question needs a model
Author	a checked model per option	validator silent
Diagnose	a health report	red check blocks
Report	estimate, mechanism, evidence	every number has a seed



The model is data, not code, so it can be checked before it runs

Static checks. A pure validator reads the model document and returns every problem with its exact path. A malformed model produces diagnostics, not a plausible wrong number. The language model proposes and a deterministic checker disposes, and the loop repeats until the checker is silent.

Nothing generated is executed. No `eval`, no generated scripts. Custom logic is one declared file.

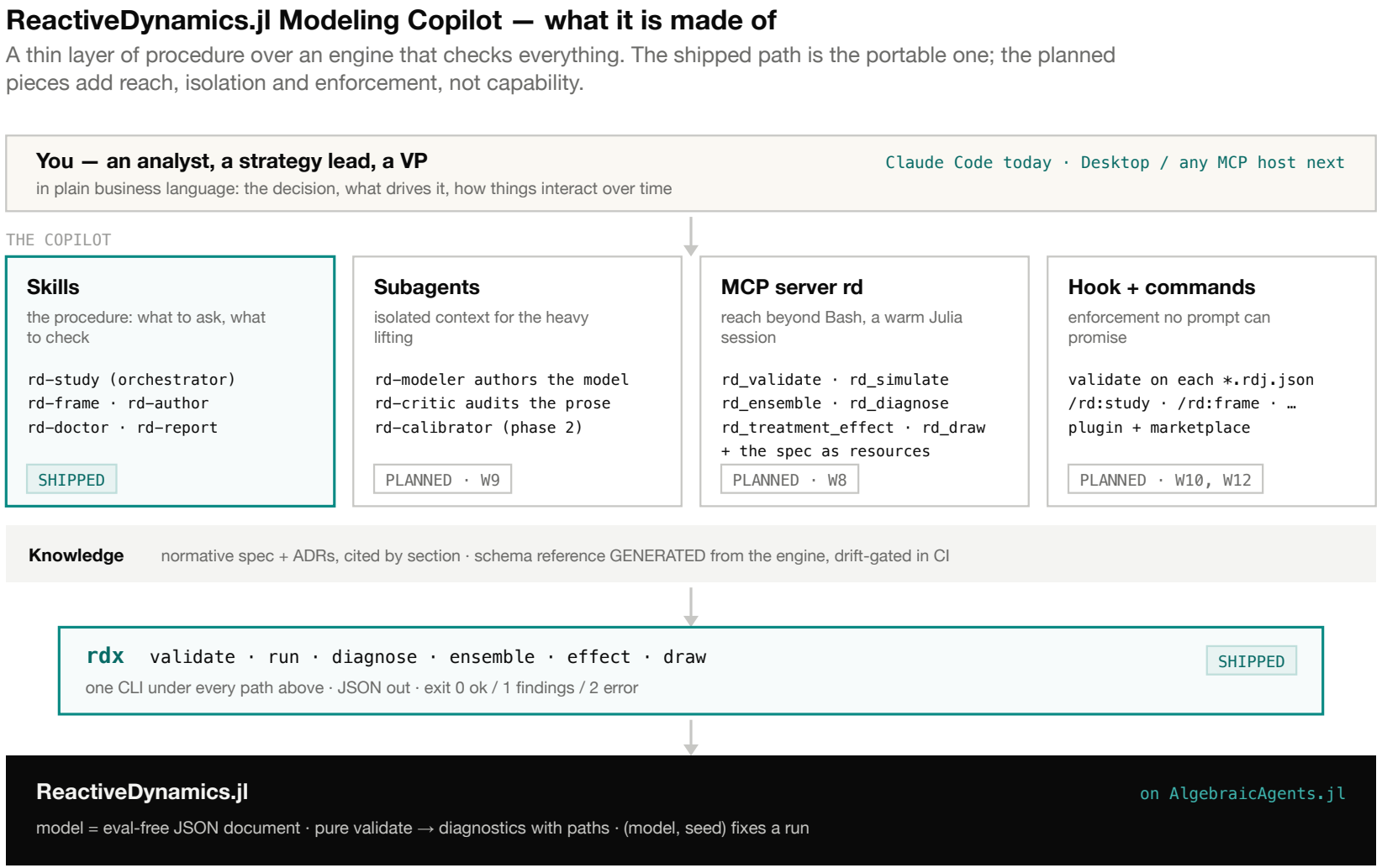
The IR. The model is a closed, eval-free document: places that hold resources, transitions that hold them over time and sometimes fail, rates that read the live state, rules that decide mid-course, costs, the arm deltas. It is diffable, serializable and replayable from `(model, seed)`.

Checked again, for meaning. Nine plausibility checks ask whether the model does what the business does: a bottleneck that never binds, a step that never fires, a horizon that cuts off the payoff.

Guarded in CI. The schema reference is generated from the engine, and a known-answer test re-derives a published result on every change.

WHAT IT IS MADE OF

A thin layer over an engine that checks everything



A REAL SESSION

Launch supply: do we pay for capacity before the readouts?

Blind-captured: two agents that cannot see each other's reasoning. The operator wrote its facts down before the first message. The company is invented, and the numbers are measurements.

Two Phase-3 biologics, one fill-finish line, a CMO offer that expires 30 November

OPERATOR

Two Phase-3 biologics are coming. […] We have one sterile fill-finish line in-house, and most of it is already taken by our marketed products. It can probably carry one launch but not two. […] Our CFO says we shouldn't pay for capacity on an asset that has less than even odds.

CLAUDE

The CMO slot works for Tern's filing, but only just. If you sign by 30 November, PPQ is done somewhere between late 2027 and early 2028. […] So the CMO can be Tern's filed site with no PAS, but only if the CMO runs on time and Tern doesn't read out early.

```
python3 -c '<in-house headroom: 12.8 batches/mo minus marketed 9/mo growing 5%/yr>'
```

One model, an arm per option, calibrated to the operator's own figures

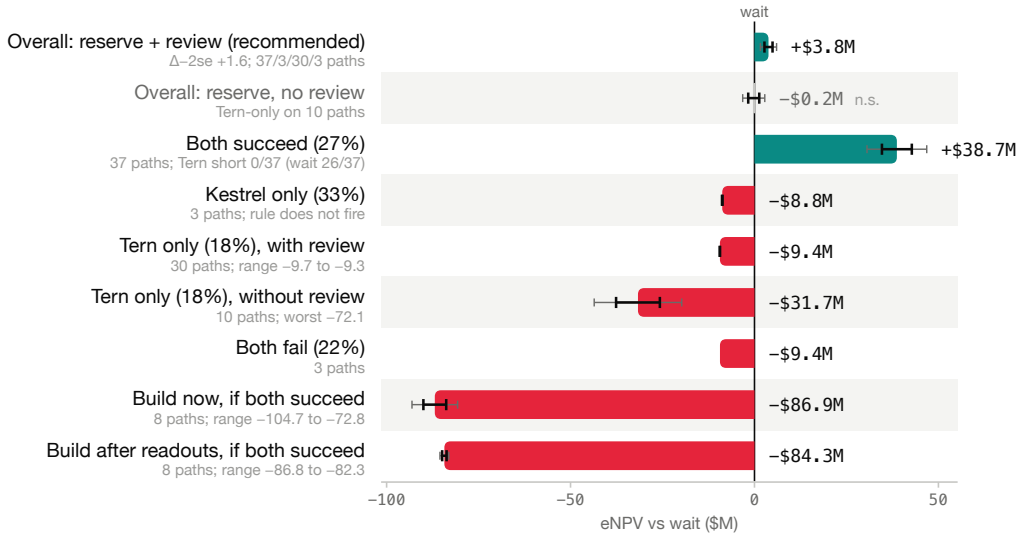
- `gen_models.py` writes one network; the arms differ only in parameters.
- `rdx validate`: 0 errors on every document.
- Calibration: Kestrel first-year revenue **$250.0M** against the operator's ~$250M, and **54** batches/yr at peak against 54.
- `rdx diagnose`: the own line binds (128 of 128 tenths used), and the design-dead branches are explained in `DOCTOR.md`.
- Options compared on **paired paths**: the same seed gives every option the same readout dates and lead times.
- Readout outcomes are forced per path and weighted by the stated odds (0.27 / 0.33 / 0.18 / 0.22).
- No option-level ensembles were run: a 96-seed ensemble took 53 minutes (W24). One wait ensemble served as a cross-check, and replays of three of its members matched exactly.

Reserve now, review at Kestrel's readout, don't build

- **+$3.8M ± $1.1M** against waiting, odds-weighted; +$25M at peak-year batch value.
- **+$39M if both succeed** (27%): Tern short at launch on 26 of 37 paths under wait, 0 of 37 reserved.
- **A bounded $9–11M cost** in every other outcome.
- **Building is $73–105M worse** even when both succeed.
- The operator's prior, a Tern-only reservation, leaves Kestrel short on every both-succeed path.

Reserve the CMO now, review at Kestrel's readout: eNPV against waiting

Paired single paths (same seed, same dates in both options), whiskers ±1 se (dark) and ±2 se (grey) across paths. Top two bars: odds-weighted overall. Others: given the outcome. Kestrel 60%, Tern 45%, 9% WACC, year-1 value per batch.



The study ships as a generated report, not a folder

Six parts, one template, every study:

1. The answer and the decision rule
2. The evidence, sensitivities and caveats
3. Why a spreadsheet cannot produce it
4. The model: elements in the operator's words, and the diff that *is* the option
5. The execution map
6. How we got it: commands, commit, the seeds

Generated, never edited. `render_report.jl` builds it from the study's own files, so it cannot disagree with the numbers.

Self-contained. One HTML file with a PDF beside it; every figure embedded, nothing to open in the repository.

Checkable. It names the commit, hashes both model documents, and lists all 208 stored runs by document and seed: any number can be replayed.

`agent/studies/launch_supply/REPORT.pdf`, 18 pages.

THE MOMENT WORTH WATCHING

The model made the tail visible, and the owner supplied the option

CLAUDE

In the other outcomes the reservation costs money: Tern only, 0.18 chance, **–$18.6M to –$72.1M**. […] That is why the expected value nets to about zero.

OPERATOR

Kestrel reads out in March 2027, nine months before Tern. […] I'd move Tern's PPQ in-house, file Tern on our own line, and give the CMO notice. That means eating the $5–10M. So the reservation isn't a fixed bet on Tern; it's something we decide to keep or drop at Kestrel's readout.

CLAUDE

Your guess holds: under the review rule, the Tern-only outcome costs $9.4M against waiting, and the $72M tail is gone.

```
python3 exit_review.py → weighted delta +3.81, se 1.10 (30 Tern-only paths)
```

HONEST ABOUT IT

What we are not claiming

- **The case is simulated.** Two role-playing agents; the operator was cooperative. It demonstrates the workflow, not anyone's supply plan.
- **Paired paths, not ensembles.** The run budget ruled ensembles out, and every figure says what it rests on.
- **The interview left soft inputs untested:** demand ±30%, marketed growth and readout correlation. The published facts audit leads with them.
- **Contention is checked end to end; feedback and coupling are not yet.** They run in the engine, and the case used feedback and a mid-course rule, but the checks do not test loops and coupled models do not go through the copilot (W25).
- **The planned components are planned.** MCP, subagents, the hook and the plugin follow endorsement.

THE ASK

Endorse the capability; we build the reach

Next	What it adds	Tracker
MCP server `rd`	the six operations as tools in any MCP host, no terminal	W8
Subagents	an isolated author and a critic that audits the prose against the numbers	W9
Validate hook	every model edit validated automatically	W10
Study card	a one-page summary over the generated report	W11
Beyond contention	checks for feedback; coupled models as documents	W25
Plugin	one-step install and slash commands	W12

The full argument, with the transcript and its evidence, is `agent/copilot.html`.